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A Comprehensive Review of Wire Electrical Discharge Machining: Principles, Parameters, Performance Measures, and Optimization Techniques

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ABSTRACT

Wire Electrical Discharge Machining (WEDM) is an advanced non-traditional machining process widely employed for precision machining of electrically conductive and difficult-to-machine materials. This review paper presents a comprehensive overview of WEDM, focusing on its working principles, process parameters, performance measures, and optimization methodologies. Special emphasis is given to machining of SS304 stainless steel, owing to its extensive industrial and biomedical applications. Various experimental, statistical, and optimization approaches such as Taguchi method, Response Surface Methodology (RSM), desirability functions, and multi-objective optimization techniques reported in the literature are critically reviewed. The paper also identifies key research gaps and future research directions in WEDM process optimization.

1. INTRODUCTION

Wire Electrical Discharge Machining (WEDM) is an advanced non-traditional machining process extensively used for precision machining of electrically conductive materials, especially those that are hard, brittle, or difficult to machine by conventional methods. In WEDM, material removal takes place due to a series of controlled electrical discharges between a continuously moving thin wire electrode and the workpiece, immersed in a dielectric medium. The localized high temperature generated by these discharges causes melting and vaporization of material, enabling the production of intricate shapes with high dimensional accuracy and superior surface quality (Kumar & Das, 2022; Rubi et al., 2024).

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With the rapid growth of the mechanical and manufacturing industries, there is an increasing demand for components made from high-strength and corrosion-resistant alloys with complex geometries and tight tolerances. Conventional machining processes such as turning, milling, and grinding often face limitations in machining such materials due to excessive tool wear, poor surface finish, and high production costs. WEDM overcomes these limitations by eliminating direct contact between the tool and the workpiece, thereby minimizing mechanical stresses and tool wear while achieving excellent precision (Alam et al., 2022; Rafeeq & Parvez, 2025).

Stainless steel SS304 is one of the most widely used austenitic stainless steels due to its excellent corrosion resistance, good mechanical strength, toughness, and biocompatibility. It finds extensive applications in aerospace, automotive, medical instruments, chemical processing equipment, and precision gauges. However, machining SS304 using conventional techniques is challenging because of its work-hardening tendency and poor machinability. WEDM has therefore emerged as a suitable alternative for machining SS304 components with complex profiles and high surface integrity (Nguyen et al., 2020; Mohapatra et al., 2024).

Despite the widespread industrial use of WEDM, achieving consistent machining quality remains a challenge due to the complex interaction of multiple process parameters such as pulse on-time, pulse off-time, peak current, servo voltage, and wire-related parameters. Improper selection of these parameters can lead to poor surface finish, dimensional inaccuracies, wire breakage, and reduced productivity (Seidi et al., 2024; Mao et al., 2025). Hence, systematic investigation and optimization of WEDM process parameters are essential to fully utilize the capabilities of the machine.

In this context, extensive research has been carried out to study the influence of WEDM process parameters on performance measures such as surface roughness, material removal rate, and dimensional accuracy. Statistical and optimization techniques such as the Taguchi method, Response Surface Methodology (RSM), grey relational analysis, and multi-objective optimization approaches have been widely adopted to identify optimal machining conditions with a reduced number of experiments (Pendokhare & Chakraborty, 2024; Afridi et al., 2025; T & Pramanik, 2024). However, limited comprehensive reviews are available that consolidate these findings, particularly for WEDM of SS304 stainless steel.

Therefore, this review paper aims to present a systematic overview of the WEDM process, its working principles, key machining parameters, performance characteristics, and optimization techniques, with special emphasis on machining of SS304 stainless steel. The review also highlights major research contributions, identifies existing research gaps, and suggests potential directions for future research in the field of WEDM.

2. **HISTORICAL DEVELOPMENT OF EDM AND WEDM**

The concept of material removal by electrical discharge dates back to the eighteenth century. In 1766, Joseph Priestley first observed the erosive effect of electrical sparks on metallic surfaces. However, for a long time, this phenomenon remained only a scientific observation without industrial application (Liao et al., 1997; Kumar & Das, 2022).

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A major breakthrough occurred during the Second World War when Boris and Natalya Lazarenko (1943) investigated the wear of tungsten electrical contacts due to sparking. Their work led to the development of the Lazarenko resistance–capacitance (RC) circuit, which laid the foundation for modern Electrical Discharge Machining (EDM). Initially, EDM was primarily used as die-sinking EDM for machining hardened steels and complex cavities in tool and die industries (Alam et al., 2022).

The evolution of Wire Electrical Discharge Machining (WEDM) began in the 1960s with the introduction of a continuously moving wire electrode in place of a solid tool electrode. The first commercially viable wire-cut EDM machine was introduced in 1967 in the Soviet Union. This innovation enabled precise machining of complex two-dimensional and three-dimensional contours with improved accuracy and reduced electrode preparation time (Rubi et al., 2024; Rafeeq & Parvez, 2025).

The advancement of numerical control (NC) and computer numerical control (CNC) technologies in the 1970s further revolutionized WEDM. By 1976, fully CNC-controlled WEDM machines became available, significantly enhancing machining accuracy, repeatability, automation, and productivity (Nguyen et al., 2020). Continuous developments in power supply systems, servo control, wire materials, and dielectric flushing techniques have since improved the stability, efficiency, and surface quality achievable by modern WEDM machines (Chen et al., 2023; Mohanraj et al., 2024).

Today, WEDM has become an indispensable manufacturing process in aerospace, automotive, electronics, medical devices, and precision tooling industries, capable of machining advanced materials with high accuracy and reliability.

3. Wire Electrical Discharge Machining (WEDM)

WEDM is a thermo-electric, non-contact machining process enabling high-precision cutting of conductive, hard-to-machine materials, with growing emphasis on multi-objective optimization and AI-assisted control.

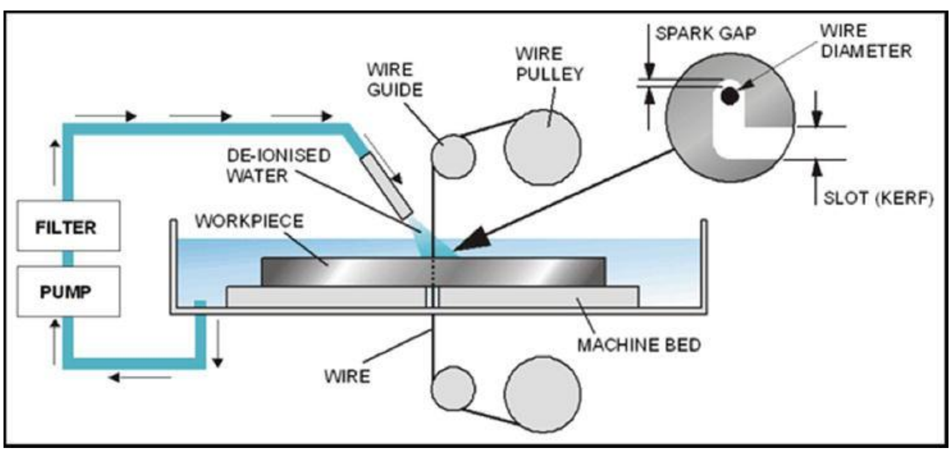

Figure 1: Key WEDM inputs and their main effects

3.1 Fundamental Principles, Parameters, and Performance

WEDM removes material via repetitive sparks between a moving wire and workpiece immersed in dielectric, eliminating cutting forces and enabling tight tolerances and complex geometries

(Afridi et al., 2025; Rubi et al., 2024; Mao et al., 2025; Alam et al., 2022). Key inputs are pulse-on/off time, peak current, gap voltage, wire feed/tension, and dielectric flow (Rubi et al., 2024; Mao et al., 2025; T & Pramanik, 2024; Alam et al., 2022). Main responses are MRR, surface roughness, kerf/width of cut, white/recast layer, wire wear, micro-defects and dimensional accuracy (Afridi et al., 2025; Rubi et al., 2024; Mao et al., 2025; Pasupuleti et al., 2025; Alam et al., 2022; Nguyen et al., 2020).

Table 1: Typical Effects of Parameters

Parameters (examples)	Dominant effects on	Citations
Ton, Toff, Ip	MRR, Ra, white layer, energy use	(Afridi et al., 2025; Pendokhare & Chakraborty, 2024; Mao et al., 2025; T & Pramanik, 2024; Alam et al., 2022; Nguyen et al., 2020)
Gap voltage, SV	Kerf, stability, wire breakage	(Afridi et al., 2025; Rubi et al., 2024; Mao et al., 2025; Pasupuleti et al., 2025; Prakash et al., 2025; Alam et al., 2022)
Wire feed/tension	Dimensional accuracy, wire wear	(Afridi et al., 2025; Rubi et al., 2024; Seidi et al., 2024; Pasupuleti et al., 2025; Alam et al., 2022)

3. 2 **Optimization Methodologies**

Traditional statistical tools (Taguchi OA, ANOVA, RSM, regression) dominate for parameter screening and response modeling (Afridi et al., 2025; Rubi et al., 2024; Pendokhare & Chakraborty, 2024; Mao et al., 2025; T & Pramanik, 2024; Pasupuleti et al., 2025; Nguyen et al., 2020; Liao et al., 1997; Nayak & Rana, 2020).

Multi-objective schemes use:

- **Desirability functions / RSM–DA** to balance MRR, Ra, taper, white layer (T & Pramanik, 2024; Nguyen et al., 2020).
- **Grey Relational Analysis and hybrid Taguchi–GRA** to condense multiple responses and identify robust settings; pulse-on time often emerges as the most influential factor (Afridi et al., 2025; Rubi et al., 2024; Seidi et al., 2024; Mao et al., 2025; Prakash et al., 2025; Nayak & Rana, 2020; Khan et al., 2023).
- **MCDM (e.g., WASPAS, MEREC)** to weight roughness, hardness, accuracy (Seidi et al., 2024).
- **Metaheuristics and swarm intelligence** (PSO, GA, GWO, Jaya, African vulture, cuckoo search, particle swarm, manta ray foraging) combined with ANN/BRNN or RSM for global multi-response optimization and intelligent control (Pendokhare & Chakraborty, 2024; Mao et al., 2025; Chen et al., 2023; Mohanraj et al., 2024; Sibalija et al., 2021; Khan et al., 2023).

3.3 Evidence Specific to Stainless Steels and SS304

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WEDM is well established for martensitic and tool steels and other stainless grades (e.g., 9Cr18MoV) using Taguchi + GRA or SNR-based multi-objective optimization to jointly increase cutting efficiency and reduce Ra with validated parameter sets (Mao et al., 2025). For **SS304**, multi-objective optimization with RBF–MOPSO and RSM–desirability achieves simultaneous reduction of Rq and white layer while increasing MRR, outperforming classical RSM–DA (Nguyen et al., 2020). Micro-pillar fabrication on SS304 shows voltage and Ton as dominant factors for kerf and feature depth, with acceptable surface integrity for functional micro-features (Mohapatra et al., 2024).

4 Research Gaps and Future Directions

Reviews emphasize needs for:

- More systematic SS304/biomedical stainless datasets covering surface integrity (residual stress, corrosion, biocompatibility) under optimized regimes (Rubi et al., 2024; Mao et al., 2025; Pasupuleti et al., 2025; Nguyen et al., 2020; Mohapatra et al., 2024).
- Integrated AI + optimization for real-time adaptive control and generalized predictive models across materials and machines (Pendokhare & Chakraborty, 2024; Mao et al., 2025; Chen et al., 2023; Sibalija et al., 2021).
- Robust studies on wire materials/coatings, hybrid cooling, and powder-mixed dielectrics in stainless steels and superalloys (Rubi et al., 2024; Kumar & Das, 2022; Rafeeq & Parvez, 2025; Pasupuleti et al., 2025; Alam et al., 2022; Mohapatra et al., 2024).

Existing work provides a strong base on WEDM fundamentals, parameter–response relationships, and multi-objective optimization, including recent advanced methods and specific SS304 studies. A literature review can build on general WEDM and optimization reviews while highlighting the relatively limited but growing body focused on SS304 surface integrity and intelligent process control.

5. Machining of SS304 Stainless Steel in WEDM

Stainless steel SS304 is one of the most widely used austenitic stainless steels due to its excellent corrosion resistance, good mechanical strength, high toughness, and biocompatibility. Because of these properties, SS304 is extensively employed in aerospace components, automotive parts, chemical processing equipment, food-processing machinery, medical instruments, and precision gauges. However, machining SS304 using conventional machining processes poses significant challenges due to its high ductility, low thermal conductivity, and strong tendency for work hardening, which often result in excessive tool wear, poor surface finish, and reduced machining efficiency.

Wire Electrical Discharge Machining (WEDM) has emerged as an effective alternative for machining SS304, particularly for components requiring complex geometries, tight tolerances, and high surface integrity. Since WEDM is a non-contact thermal erosion process, it eliminates cutting forces and tool wear, making it suitable for machining SS304 irrespective of its hardness or strength. The material removal mechanism in WEDM involves localized melting and vaporization of SS304 caused by controlled electrical discharges between the wire electrode and the workpiece in the presence of a dielectric medium.

Several studies reported in the literature indicate that the machining performance of SS304 in WEDM is strongly influenced by key process parameters such as pulse on-time, pulse off-time, peak current, servo voltage, wire feed rate, and wire tension. Among these, pulse on-time and peak current play a dominant role in determining material removal rate and surface roughness. An increase in pulse on-time or peak current generally enhances the material removal rate due to higher discharge energy; however, it also leads to deeper and irregular craters on the machined surface, resulting in increased surface roughness and the formation of a recast layer.

Pulse off-time is equally important in machining SS304, as it governs the deionization of the dielectric fluid and effective flushing of molten debris from the spark gap. Higher pulse off-time improves surface finish and process stability by reducing arcing and wire breakage, although it may reduce productivity. Proper selection of pulse off-time is therefore critical to achieving a balance between surface quality and machining speed.

Surface integrity is a key concern in WEDM of SS304, especially for applications involving fatigue loading or biomedical use. Studies have shown that improper parameter selection can result in micro-cracks, residual tensile stresses, and the formation of a heat-affected zone (HAZ). Optimization of process parameters using statistical techniques such as the Taguchi method, Response Surface Methodology (RSM), and desirability-based multi-objective optimization has been shown to significantly improve surface roughness and dimensional accuracy while maintaining acceptable material removal rates.

Despite considerable research on WEDM of various steels, limited studies specifically address comprehensive modeling and multi-objective optimization of surface roughness and surface integrity in SS304 using different wire electrode materials. This highlights the need for further systematic investigations focusing on the combined effects of electrical, mechanical, and thermal parameters during WEDM of SS304 stainless steel.

Overall, WEDM has proven to be a highly suitable and reliable machining process for SS304 stainless steel, provided that machining parameters are carefully selected and optimized to meet specific industrial requirements related to surface quality, accuracy, and productivity.

6. Research Gaps and Future Scope

A critical review of the existing literature on Wire Electrical Discharge Machining (WEDM) reveals that significant progress has been made in understanding the fundamental mechanisms, process parameters, and optimization techniques for machining various conductive materials. However, several research gaps still exist, particularly with respect to machining of SS304 stainless steel.

Most of the reported studies focus primarily on individual performance characteristics such as material removal rate or surface roughness, whereas limited work has been carried out on comprehensive multi-objective optimization involving surface integrity, dimensional accuracy, residual stresses, and recast layer thickness simultaneously. In addition, although pulse on-time, pulse off-time, and peak current have been widely investigated, the combined influence of electrical parameters with wire-related parameters such as wire material, wire tension, and wire feed rate in WEDM of SS304 remains insufficiently explored.

Another major gap identified is the limited consideration of thermo-physical properties of SS304 in process modeling. Most empirical models developed in earlier studies rely solely on machining parameters and neglect the influence of material properties, which restricts their general applicability. Furthermore, very few studies have addressed real-time monitoring, adaptive control, and intelligent optimization techniques such as artificial intelligence, machine learning, and hybrid optimization algorithms for WEDM of stainless steels.

From an industrial perspective, studies on surface integrity aspects such as fatigue life, corrosion behavior, and biocompatibility of WEDM-machined SS304 components are scarce, despite their importance in aerospace and biomedical applications. Environmental and sustainability aspects, including energy consumption, dielectric recycling, and green machining practices in WEDM, also remain underexplored.

Future research should therefore focus on developing integrated experimental–computational models that combine machining parameters with material properties. Advanced optimization techniques, real-time monitoring systems, and AI-based adaptive control strategies can be employed to enhance machining efficiency and consistency. In addition, systematic investigations on surface integrity, functional performance, and sustainable machining of SS304 using WEDM will significantly broaden its industrial applicability.

7. Conclusion

This review paper has presented a comprehensive overview of Wire Electrical Discharge Machining (WEDM), covering its historical development, working principles, process parameters, optimization techniques, and applications, with special emphasis on machining of SS304 stainless steel. The review highlights that WEDM is a highly effective non-traditional machining process capable of producing complex geometries with high dimensional accuracy and acceptable surface quality, especially for materials that are difficult to machine using conventional methods.

The analysis of literature indicates that key process parameters such as pulse on-time, pulse off-time, and peak current play a dominant role in controlling material removal rate and surface roughness during WEDM of SS304. Statistical and optimization techniques such as the Taguchi method, Response Surface Methodology, and multi-objective optimization approaches have been successfully employed to determine optimal machining conditions while minimizing experimental effort.

Despite the extensive research reported, several gaps remain, particularly in the areas of integrated modeling, surface integrity analysis, intelligent control, and sustainability. Addressing these gaps through advanced experimental and analytical approaches will further enhance the understanding and industrial implementation of WEDM for SS304 stainless steel.

In conclusion, WEDM holds significant potential for precision machining of SS304 stainless steel, and continued research in process optimization, surface integrity, and smart manufacturing will contribute to improved performance, reliability, and wider industrial adoption of this versatile machining technology.

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8. Conflict of Interest

The author(s) declare that there is **no conflict of interest** regarding the publication of this review paper.

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